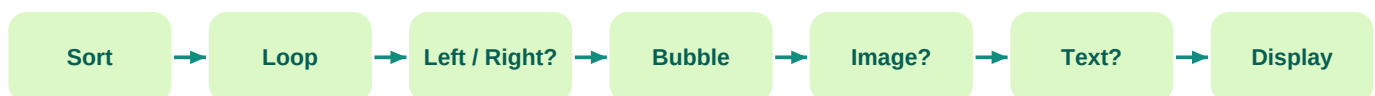


Let's understand it visually — like a WhatsApp chat. Suppose your **messages** array contains the following three items:

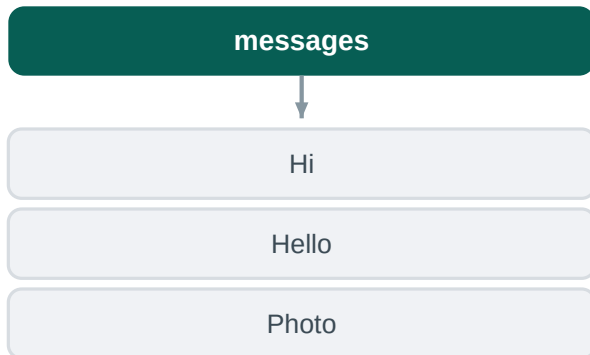
```
[
  {
    text: "Hi",
    to_user_id: "user2",
    message_type: "text"
  },
  {
    text: "Hello",
    to_user_id: "myId",
    message_type: "text"
  },
  {
    text: "Photo",
    to_user_id: "user2",
    message_type: "image",
    media_url: "photo.jpg"
  }
]
```

This guide breaks the rendering logic into seven simple steps — from sorting the raw array, to deciding whether a bubble sits on the left or right, to finally showing text or an image inside it. Each step is shown as a small diagram so the flow is easy to follow.



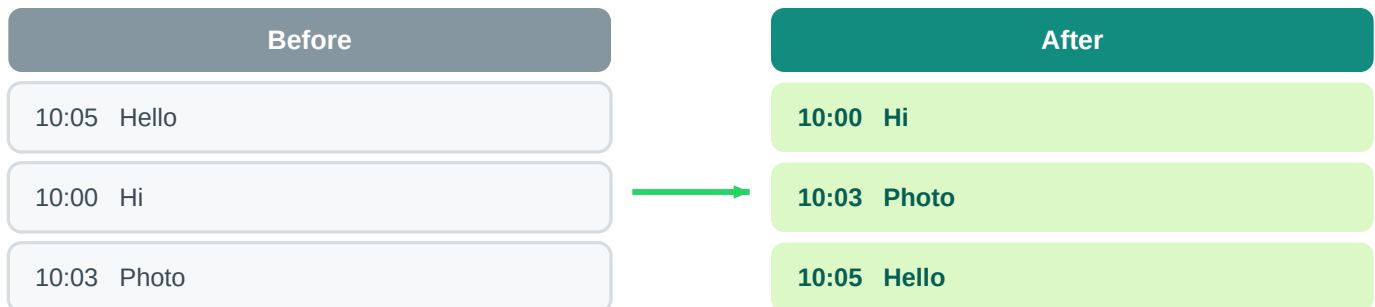
1 The Raw Message List

Before anything is rendered, the messages simply exist as an unsorted list.



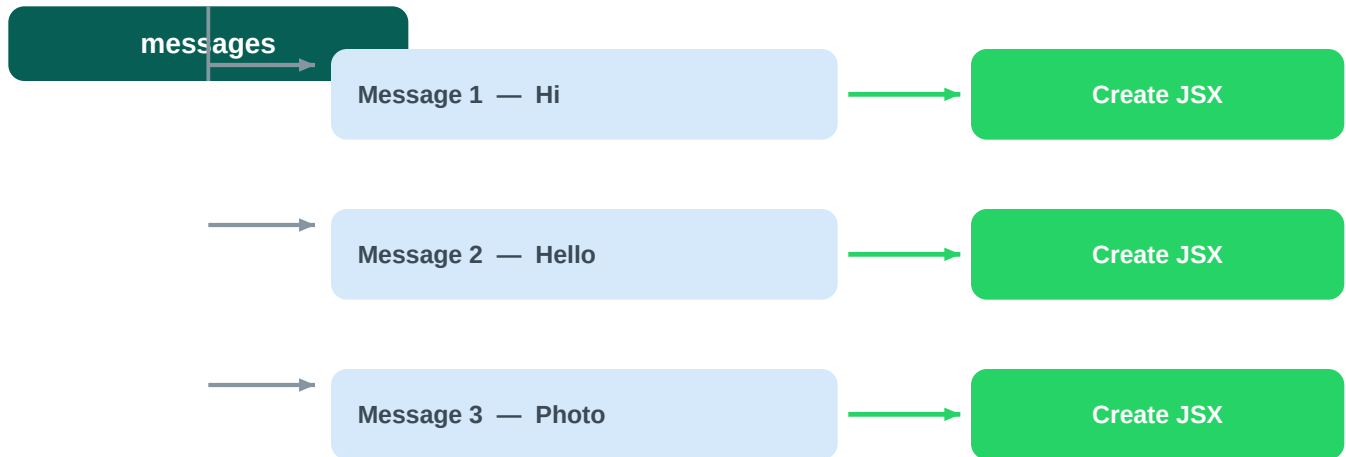
2 Sort by Time (.toSorted)

The messages are re-ordered chronologically before display.



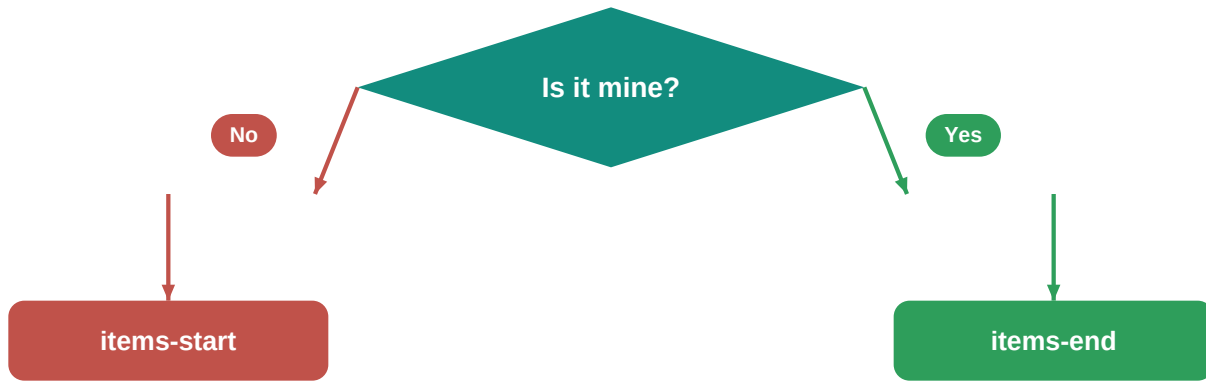
3 Looping Through Every Message (.map)

React walks through the sorted array one message at a time and turns each one into a piece of JSX.

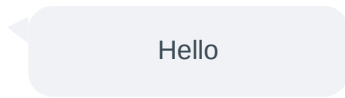


4 Is This Message Mine?

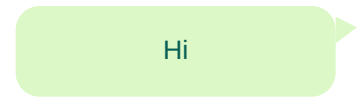
React checks `message.to_user_id !== user._id` to decide which side of the screen the bubble belongs on.



Someone else's message — starts from the left



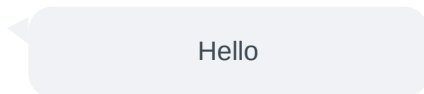
My message — moves to the right



5 Creating the Bubble

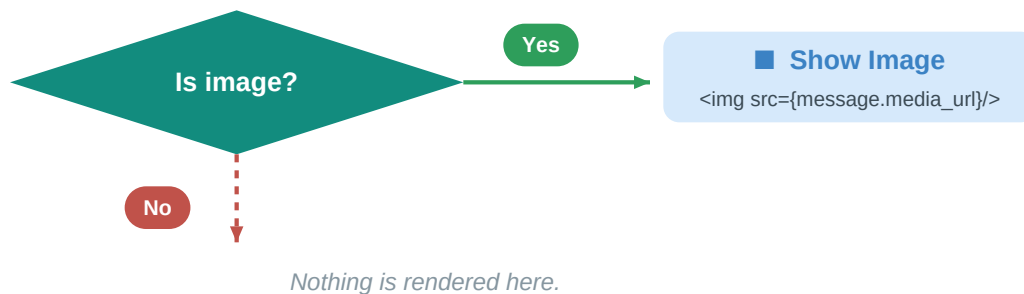
Once the side is decided, React wraps the message in a rounded, shadowed container.

```
<div className="p-2 rounded-lg shadow">
```



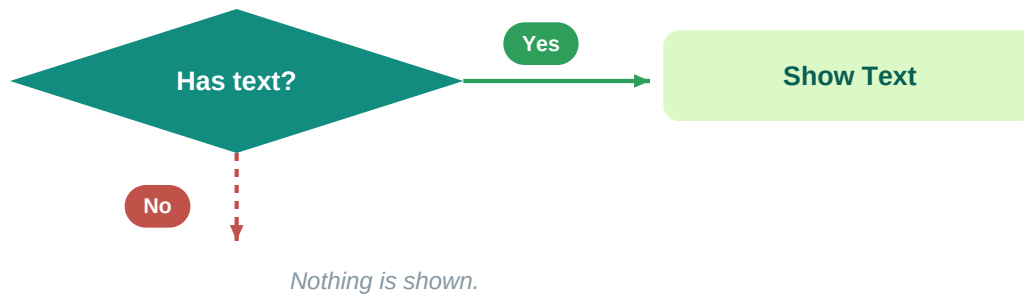
6 Is It an Image?

React checks `message.message_type === "image"` before deciding what to render inside the bubble.



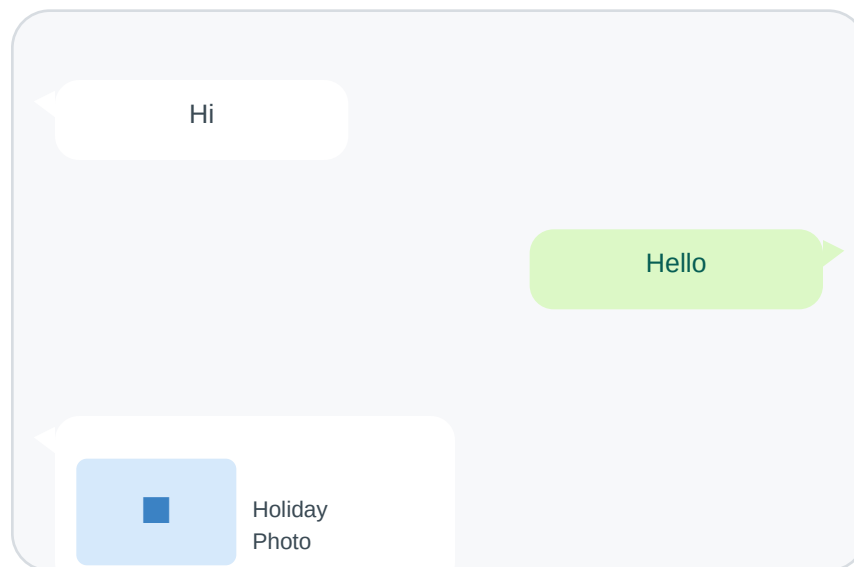
7 Does Text Exist?

Finally, React checks `message.text` & `<p>{message.text}</p>` before printing any caption or message text.

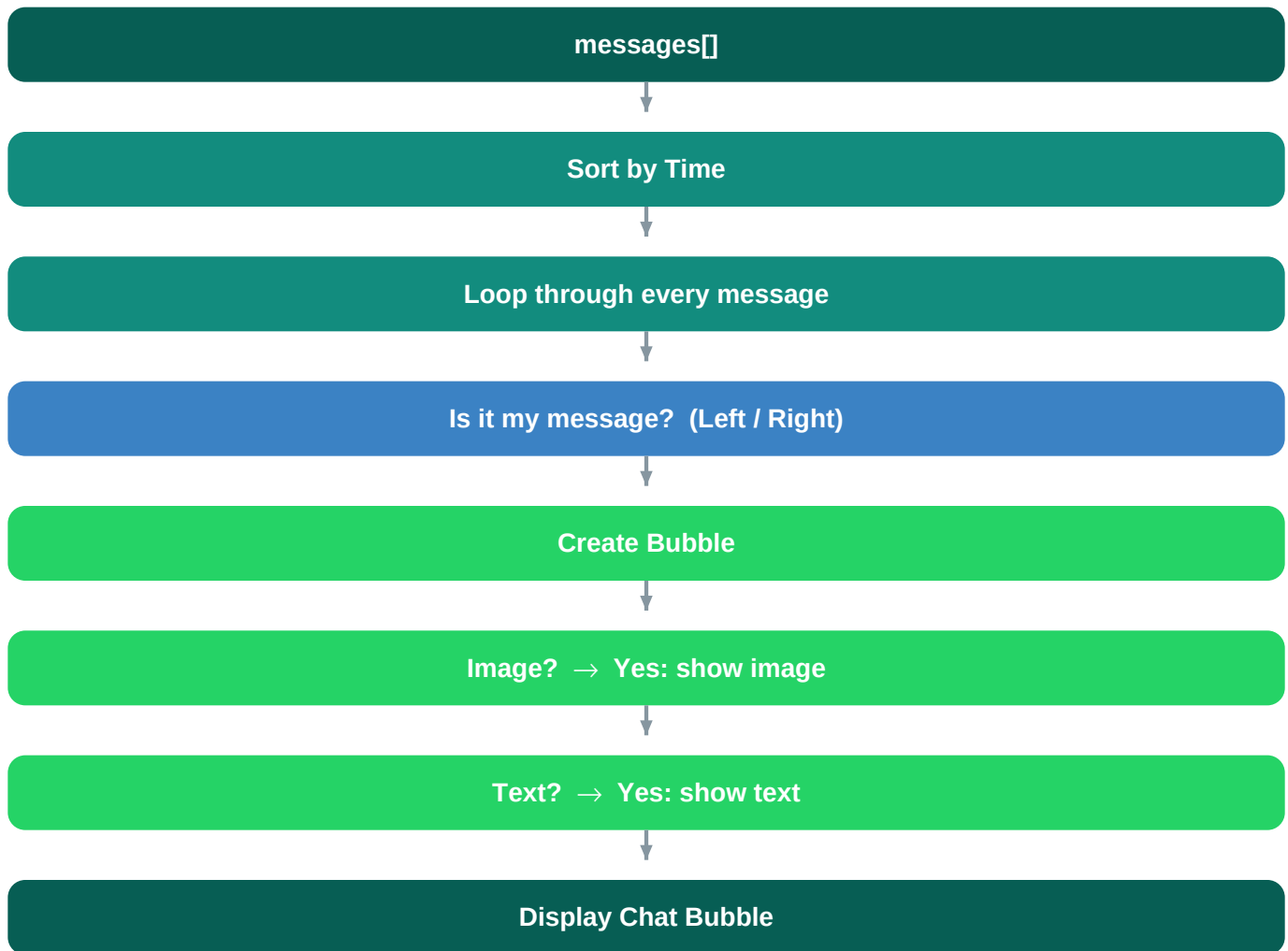


Final Output

Putting all seven steps together for the three sample messages:



The Entire Flow, End to End



This is exactly how React builds a chat screen: take each message → decide left or right → create a bubble → optionally show an image → show the text → move to the next message.